# Computer Science & IT



## Discrete Mathematics

## Set Theory & Algebra

**Lecture No. 01**



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# Topics to be Covered



- Topic: Sets and representation of sets
- Topic: Different types of sets
- Topic: [redacted]
- Topic: [redacted]

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# Topic : Syllabus

Marks

↓

Engg Maths + Discrete Maths

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13 Marks

(7 - 9)

1. Set Theory & Algebra
   - Sets & Relations
   - Functions ✓
   - Group Theory ✓
2. Graph Theory
3. Mathematical Logic
   - Propositional logic ✓
   - Predicate Logic ✓
4. Combinatorics
   - Recurrence relation
   - Generating function

↓

Basic of Combinatorics

Engg Maths.

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Pre-requisite:

"P&C" 11th Class.



# Topic : Set

- A well-defined unordered collection of distinct elements is called as set.

$A = \{1, 2, 3, 4, Jan, Mon\} = \{Jan, 1, 2, 3, 4, Mon\}$

$B = \{1, 2, 3, 4, 5\}$

$C = \{a, e, i, o, u\}$

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# Topic : Representation of sets

- There are three different ways in which set may be represented.

1. Roaster form or Tabular form:
2. Set-builder form:
3. Statement form:



# Topic : Roaster form or Tabular form

In this notation we will list all the elements of the set within the curly braces.

eg: $A = \{1, 2, 3, 4, 5\}$

$B = \{1, 2, 3, Jan, feb\}$ { We may not represent the elements of set B using "set-builder" notation }

$X = \{1, 2, 3, 4, \ldots\ldots\}$ = Set of all Natural Numbers

$D = \{0, 0.25, 0.3, 0.31, 0.4, 0.79, 0.8, 0.9\}$

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# Topic : Set-builder form

* In this notation we define the property and all the elements satisfying that property will be members of the set

$$A = \{ x \mid x \in N \text{ and } x \leq 5 \} = \{1, 2, 3, 4, 5\}$$

- $A$ = it is a set
- $x$ = containing elements $x$
- such that
- $x \in N$ and $x \leq 5$ = Property

$$C = \{ x : x \in R \text{ and } 0 \leq x \leq 1 \}$$

- $x$ = such that
- $R$ = Real number

We can not represent set C in roaster form

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# Topic : Statement form

A = Set of all natural numbers less than or equal to 5.

C = Set of all real numbers between 0 & 1

Slide



# Topic : Cardinality of a set

{ Cardinality is generally defined for finite sets }

Cardinality of a set A is denoted by $|A|$ and $|A|$ represent the number of elements in set A.

eg, $A = \{1, 2, 3, Jan, Feb\}$

$|A| = 5$

Slide



# Topic : Types of sets

**Empty Set:** A set with no element in it is called an Empty set.

Empty set can be denoted by $\{\}$ or $\emptyset$

$$\{\} = \emptyset$$

$|\{\}| = |\emptyset| = 0$

$|\{1, \{2,3\}\}| = 2$

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# Topic : Types of sets

**Singleton Set:** A set with exactly '1' element in it is called a Singleton set

eg $A = \{a\}$

$\{1\}$

$\{\{2, \{3,4\}\}\}$

$\{\{1, \{1,2\}\}\}$

$\{\{\}\}$

its a set

Containing one element & that element is an empty set

Slide

$\{\} = \emptyset$

$\therefore \quad |\{\}| = |\emptyset| = 0$

$\{\}$ : it is an Empty set

$\{\emptyset\} = \{\{\}\}$ : It is a set Containing one element that Element is an Empty set

$|\{\}| = 0$ but $|\{\{\}\}| = 1$

$$\{\} \neq \{\{\}\}$$



# Topic : Types of sets

**Finite Set:** A set that contains finite number of elements is called a finite set

eg: $\{1\}$, $\{1,2\}$, $\{a,b,c\}$ etc..

$\{\ \}$

⟵ Empty set is always a finite set.

Slide



# Topic : Types of sets

**Infinite Set:** A set that contains infinite number of elements in it is called an infinite set.

Eg: A = Set of all natural numbers

B = Set of all real numbers

C = Set of all Even integers

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Another Classification w.r.t. number of elements in the set

- Finite sets
  - Finite sets are always Countable
- Infinite sets
  - Countably infinite sets
  - Uncountably infinite sets

# THANK - YOU

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